

As such, these fields (basic certificate fields "TbsCertificate", "SignatureAlgorithm" and "SignatureValue") are not marked as mandatory or critical in the certificate profiles below.

[V2G20-2694] The relying party shall reject a certificate that does not contain either "TbsCertificate", "SignatureAlgorithm" or "SignatureValue".

Per ITU-T X.509 and IETF RFC 5280 (as updated by IETF RFC 6818, IETF RFC 8398 and IETF RFC 8399), "TbsCertificate" field will always contain sub-fields "Version", "SerialNumber", "Signature", "Issuer", "Validity", "Subject" and "SubjectPublicKeyInfo". If any of these fields are not present in the certificate or if the relying party/validating entity is unable to process any of these sub-fields, the relying party will reject the certificate.

As such, these parameters ("Version", "SerialNumber", "Signature", "Issuer", "Validity", "Subject" and "SubjectPublicKeyInfo") are not marked as mandatory or critical in the certificate profiles below.

[V2G20-2695] The relying party shall reject a certificate that does not contain either "Version", "SerialNumber", "Signature", "Issuer", "Validity", "Subject" or "SubjectPublicKeyInfo" in "TbsCertificate" field.

[V2G20-2696] Each relying party shall be capable of interpreting/processing "AuthorityKeyIdentifier", "SubjectKeyIdentifier", "ExtendedKeyUsage", "CRLDistributionPoints", "AuthorityInfoAccess" and "SubjectInfoAccess" extensions.

This document does not require relying party to be capable of interpreting/processing CertificatePolicies extension. It is left to the local policy of the relying party whether it needs to process the CertificatePolicies extension.

[V2G20-1851] keyIdentifier in "AuthorityKeyIdentifier" shall be calculated as in IETF RFC 5280:2008, 4.2.1.2, method 2 using the public key of the issuer and shall be included as OCTET STRING in the certificate.

NOTE 1 IETF RFC5280 has been updated by IETF RFC 6818, IETF RFC 8398 and IETF RFC 8399. These updates are considered to be included in this document.

Per IETF RFC 5280 (as updated by IETF RFC 6818, IETF RFC 8398 and IETF RFC 8399) and ITU-T X.509, it is optional to include "authorityCertIssuer" and "authorityCertSerialNumber" in "AuthorityKeyIdentifier". But if "authorityCertIssuer" or "authorityCertSerialNumber" is present, then it is mandatory to include the corresponding "authorityCertSerialNumber"/"authorityCertIssuer" in the order shown. If "authorityCertIssuer" is present while "authorityCertSerialNumber" is not present or if "authorityCertSerialNumber" is present while "authorityCertIssuer" is not present, the relying party will reject the certificate.

[V2G20-1852] keyIdentifier in "SubjectKeyIdentifier" shall be calculated as IETF RFC 5280:2008, 4.2.1.2, method 2 using the public key of its own certificate and shall be included as OCTET STRING in the certificate.

NOTE 2 IETF RFC 5280 has been updated by IETF RFC 6818, IETF RFC 8398 and IETF RFC 8399. These updates are considered to be included in this document.

[V2G20-2582] Any URIs in the certificates (for example OCSP address, etc.) shall conform to IETF RFC 3986 as updated by IETF RFC 6874 and IETF RFC 7320.

[V2G20-1224] The distinguished name shall be represented as a string that complies with IETF RFC 4514, Clause 3. Per IETF RFC 5280 (as updated by IETF RFC 6818, IETF RFC 8398 and IETF RFC 8399), any certificate policy shall be valid for entire

certification path. This means the "certificatePolicies" extension shall appear in all certificates in the certificate chain except the root certificate. Certificate policy is provided in the certificates using the OIDs. Hence, the certificate policy OID provided in a leaf certificate shall also appear in the sub-CA1 (if present) and sub-CA2 certificates to be valid for the entire certification path.

For example, if the "certificatePolicies" extension is missing from sub-CA1 certificate, no explicit certificate policies are allowed below that CA certificate.

"certificatePolicies" extension should not appear in root certificates as "certificatePolicies" is implicitly set to "anyPolicy" in the root certificate. If the "certificatePolicies" extension were to be present in a root certificate, any updates/changes/modification to the policies would require generating a new root certificate.

If "CRLDistributionPoints" is present, "distributionPoint", "reasons" and "cRLIssuer" that are part of "CRLDistributionPoints" may or may not be present while "CRLDistributionPoints" is present.

Per IETF RFC 5280 (as updated by IETF RFC 6818, IETF RFC 8398 and IETF RFC 8399), if "AuthorityInfoAccess" is present, "accessMethod" and "accessLocation" that are part of "AuthorityInfoAccess" are also present. If "accessMethod" and "accessLocation" are not present while "AuthorityInfoAccess" is present, the relying party will reject the certificate.

Per IETF RFC 5280 (as updated by IETF RFC 6818, IETF RFC 8398 and IETF RFC 8399), if "SubjectInfoAccess" is present, "accessMethod" and "accessLocation" that are part of "SubjectInfoAccess" are also present. If "accessMethod" and "accessLocation" are not present while "SubjectInfoAccess" is present, the relying party will reject the certificate.

B.3 V2G root CA certificate profiles

B.3.1 Based on curves as defined by [V2G20-2674]

Table B.3 outlines the V2G root CA certificate based on curves as defined by [V2G20-2674].

Table B.3 — V2G root CA certificate based on curves as defined by [V2G20-2674]

ISO 15118-20 certificate profiles			V2G root V2G
TbsCertificate	Version		2 (X.509v3)
	SerialNumber		Integer
	Signature	AlgorithmIdentifier	x / c
		algorithm	x / c ecdsa-with-SHA512
	Issuer	Country	(x)
		Organization	x
		Organization unit	(x)
		Common name	x
		Domain component	(x)
	Validity	Validity	x (25 years)
		notBefore	x
		time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]

ISO 15118-20 certificate profiles		V2G root V2G
	notAfter	x
	time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
	Subject	Country (x)
	Organization	x
	Organization unit	(x)
	Common name	x
	Domain component	(x)
	SubjectPublicKeyInfo	AlgorithmIdentifier x algorithm id-ecPublicKey parameters x ECParameters namedCurve x secp521r1 SubjectPublicKey x (BIT STRING)
	IssuerUniqueID	-
	SubjectUniqueID	-
	AuthorityKeyIdentifier	-
	SubjectKeyIdentifier	x / nc id-ce-subjectKeyIdentifier
	keyIdentifier	x (see [V2G20-1852])
	KeyUsage	x / c id-ce-keyUsage
	digitalSignature	0/1
	nonRepudiation (contentCommitment)	0/1
	keyEncipherment	0/1
	dataEncipherment	0
	keyAgreement	0/1
	keyCertSign	1
	cRLSign	0/1
	encipherOnly	0
	decipherOnly	0
	ExtendedKeyUsage	-
	CertificatePolicies	-
	BasicConstraints	x / c id-ce-basicConstraints
	CA	true
	PathLength	-
	CRLDistributionPoints (CRL)	-
	AuthorityInfoAccess (OCSP)	-
SignatureAlgorithm	AlgorithmIdentifier	x
	algorithm	x ecdsa-with-SHA512
	parameters	-
SignatureValue		(Octet-String)

NOTE 1 ECPParameters is simply a container that contains the parameter values for the selected algorithm.

NOTE 2 OID values for "ecdsa-with-SHA512", "id-ecPublicKey" and "secp521r1" can be found in B.1.

B.3.2 Based on curves as defined by [V2G20-2319]

Table B.4 outlines the V2G root CA certificate based on curves as defined by [V2G20-2319].

Table B.4 — V2G root CA certificate based on curves as defined by [V2G20-2319]

ISO 15118-20 certificate profiles		V2G root V2G
TbsCertificate	Version	2 ("2" indicates X.509v3)
	SerialNumber	(Positive integer)
	Signature	AlgorithmIdentifier
		algorithm
		parameters
	Issuer	Country
		Organization
		Organization unit
		Common name
		Domain component
	Validity	Validity
		notBefore
		time
		notAfter
		time
		time
	Subject	Country
		Organization
		Organization unit
		Common name
		Domain component
	SubjectPublicKeyInfo	AlgorithmIdentifier
		algorithm
		parameters
		SubjectPublicKey
	IssuerUniqueID	
	SubjectUniqueID	
	Extensions	AuthorityKeyIdentifier
		SubjectKeyIdentifier
		keyIdentifier
		KeyUsage
		digitalSignature
		nonRepudiation (contentCommitment)
		keyEncipherment
		dataEncipherment
		keyAgreement
		keyCertSign
		cRLSign
		encipherOnly
		decipherOnly
		ExtendedKeyUsage

ISO 15118-20 certificate profiles			V2G root V2G
		CertificatePolicies	-
		BasicConstraints	x / c id-ce-basicConstraints
		CA	true
		PathLength	-
		CRLDistributionPoints (CRL)	-
		AuthorityInfoAccess (OCSP)	-
SignatureAlgo rithm	AlgorithmIdentifier		x
	algorithm		x id-Ed448
	parameters		-
SignatureValue			(Raw BIT STRING)

B.4 Charge point operator certificate profiles

B.4.1 Based on curves as defined by [V2G20-2674]

Table B.5 outlines the charge point operator certificates based on curves as defined by [V2G20-2674].

Table B.5 — Charge point operator certificates based on curves as defined by [V2G20-2674]

ISO 15118-20 certificate profiles			CSO sub-CA1 Sub	Charge point operator CSO sub-CA2 Sub	SECC certificate Leaf
TbsCertificate	Version		2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)
	SerialNumber		(Positive integer)	(Positive integer)	(Positive integer)
	Signature	AlgorithmIdentifier	x	x	x
		algorithm	x ecdsa-with-SHA512	x ecdsa-with-SHA512	x ecdsa-with-SHA512
		parameters	-	-	-
	Issuer	Country	(x)	(x)	(x)
		Organization	x	x	x
		Organization unit	(x)	(x)	(x)
		Common name	x	x	x
		Domain component	(x)	(x)	(x)
	Validity	Validity	x [up to CSO]	x [up to CSO]	x [up to CSO]
		notBefore	x	x	x
		time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
		notAfter	x	x	x
		time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
		Country	(x)	(x)	x
		Organization	x	x	x
		Organization unit	(x)	(x)	(x)

ISO 15118-20 certificate profiles		CSO sub-CA1 Sub	Charge point operator CSO sub-CA2 Sub	SECC certificate Leaf
		Common name	x	x SECCID (see [V2G20-2583])
		Domain component	(x)	(x) & "CSO"
	SubjectPublicKeyInfo	AlgorithmIdentifier	x	x
		algorithm	x id-ecPublicKey	x id-ecPublicKey
		parameters	x ECParameters	x ECParameters
		namedCurve	xsecp521r1	x secp521r1
		SubjectPublicKey	x (BIT STRING)	x (BIT STRING)
	IssuerUniqueID		-	-
	SubjectUniqueID		-	-
	Extensions	AuthorityKeyIdentifier	x / nc id-ce- authorityKeyIdentifier	x / nc id-ce- authorityKeyIdentifier
		keyIdentifier	x (see [V2G20-1851])	x (see [V2G20-1851])
		authorityCertIssuer	(x)	(x)
		authorityCertSerial Number	(x) (Positive integer)	(x) (Positive integer)
		SubjectKeyIdentifier	x / nc id-ce-subjectKeyIdentifier	x / nc id-ce- subjectKeyIdentifier
		keyIdentifier	x (see [V2G20-1852])	x (see [V2G20-1852])
		KeyUsage	x / c id-ce-keyUsage	x / c id-ce-keyUsage
		digitalSignature	0/1	1
		nonRepudiation (contentCommitment)	0/1	0/1
		keyEncipherment	0/1	0/1
		dataEncipherment	0	0
		keyAgreement	0/1	1
		keyCertSign	1	0
		cRLSign	1	0
		encipherOnly	0	0
		decipherOnly	0	0
		ExtendedKeyUsage	-	(x) / c id-ce-extKeyUsage
		keyPurposeId	-	x id-kp-serverAuth
		CertificatePolicies	(x) / nc	(x) / nc
		BasicConstraints	x / c id-ce-basicConstraints	x / c id-ce-basicConstraints
		CA	true	false
		PathLength	1	-
		CRLDistributionPoints (CRL)	-	-
		AuthorityInfoAccess (OCSP)	x / c id-pe-authorityInfoAccess	x / c id-pe- authorityInfoAccess
		accessMethod	x id-ad-ocsp	x id-ad-ocsp
		accessLocation	x (location of the OCSP responder)	x (location of the OCSP responder)
SignatureAlgorithm	AlgorithmIdentifier		x	x
	algorithm		x ecdsa-with-SHA512	x ecdsa-with-SHA512
	parameters		-	-
SignatureValue		(Octet-String)	(Octet-String)	(Octet-String)

NOTE 1 ECPParameters is simply a container that contains the parameter values for the selected algorithm.

NOTE 2 OID values for "ecdsa-with-SHA512", "id-ecPublicKey" and "secp521r1" can be found in B.1.

Although including "keyPurposeId" (as part of "ExtendedKeyUsage") is mentioned as optional, it is highly recommended to include this parameter in the SECC certificate. This makes the certificate more secure and prevents its misuse.

B.4.2 Based on curves as defined by [V2G20-2319]

Table B.6 outlines the charge point operator certificates based on curves as defined by [V2G20-2319].

Table B.6 — Charge point operator certificates based on curves as defined by [V2G20-2319]

ISO 15118-20 certificate profiles		CSO sub-CA1 Sub	Charge point operator CSO sub-CA2 Sub	SECC certificate Leaf
TbsCertificate	Version	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)
	SerialNumber	(Positive integer)	(Positive integer)	(Positive integer)
	Signature	AlgorithmIdentifier	x	x
		algorithm	x id-Ed448	x id-Ed448
		parameters	-	-
	Issuer	Country	(x)	(x)
		Organization	x	x
		Organization unit	(x)	(x)
		Common name	x	x
		Domain component	(x)	(x)
	Validity	Validity	x [up to CSO]	x [up to CSO]
		notBefore	x	x
		time	x	x
			(GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	(GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
		notAfter	x	x
		time	x	x
			(GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	(GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
	Subject	Country	(x)	x
		Organization	x	x
		Organization unit	(x)	(x)
		Common name	x	x SECCID (see [V2G20-2583])
		Domain component	(x)	(x) & "CSO"
	SubjectPublic KeyInfo	AlgorithmIdentifier	x	x
		algorithm	x id-Ed448	x id-Ed448
		parameters	-	-
		SubjectPublicKey	x (BIT STRING)	x (BIT STRING)
	IssuerUniqueID	-	-	-
	SubjectUniqueID	-	-	-

ISO 15118-20 certificate profiles		CSO sub-CA1 Sub	Charge point operator CSO sub-CA2 Sub	SECC certificate Leaf
Extensions	AuthorityKeyIdentifier	x / nc id-ce- authorityKeyIdentifier	x / nc id-ce- authorityKeyIdentifier	x / nc id-ce- authorityKeyIdentifier
	keyIdentifier	x (see [V2G20-1851])	x (see [V2G20-1851])	x (see [V2G20-1851])
	authorityCertIssuer	(x)	(x)	(x)
	authorityCertSerial Number	(x) (Positive integer)	(x) (Positive integer)	(x) (Positive integer)
	SubjectKeyIdentifier	x / nc id-ce-subjectKeyIdentifier	x / nc id-ce-subjectKeyIdentifier	x / nc id-ce-subjectKeyIdentifier
	keyIdentifier	x (see [V2G20-1852])	x (see [V2G20-1852])	x (see [V2G20-1852])
	KeyUsage	x / c id-ce-keyUsage	x / c id-ce-keyUsage	x / c id-ce-keyUsage
	digitalSignature	0/1	0/1	1
	nonRepudiation (contentCommitme nt)	0/1	0/1	0/1
	keyEncipherment	0/1	0/1	0/1
	dataEncipherment	0	0	0
	keyAgreement	0/1	0/1	1
	keyCertSign	1	1	0
	cRLSign	1	1	0
	encipherOnly	0	0	0
	decipherOnly	0	0	0
	ExtendedKeyUsage	-	-	(x) / c id-ce-extKeyUsage
	keyPurposeId	-	-	x id-kp-serverAuth
	CertificatePolicies	(x) / nc	(x) / nc	(x) / nc
	BasicConstraints	x / c id-ce-basicConstraints	x / c id-ce-basicConstraints	x / c id-ce-basicConstraints
	CA	true	true	false
	PathLength	1	0	-
	CRLDistributionPoints (CRL)	-	-	-
	AuthorityInfoAccess (OCSP)	x / c id-pe-authorityInfoAccess	x / c id-pe-authorityInfoAccess	x / c id-pe-authorityInfoAccess
	accessMethod	x id-ad-ocsp	x id-ad-ocsp	x id-ad-ocsp
	accessLocation	x (location of the OCSP responder)	x (location of the OCSP responder)	x (location of the OCSP responder)
SignatureAlg orithm	AlgorithmIdentifier	x	x	x
	algorithm	x id-Ed448	x id-Ed448	x id-Ed448
	parameters	-	-	-
SignatureValue		(Raw BIT STRING)	(Raw BIT STRING)	(Raw BIT STRING)

Although including "keyPurposeId" (as part of "ExtendedKeyUsage") is mentioned as optional, it is highly recommended to include this parameter in the SECC certificate. This makes the certificate more secure and prevents its misuse.

B.4.3 Common requirements for charge point operator certificate profiles

As mentioned in Table B.5 and Table B.6, it is optional to include the "certificatePolicies" extension in any of the charge point operator certificates. If the "certificatePolicies" extension is included, it can be marked as non-critical so that the relying party can ignore this extension if it cannot process it. This is done to improve the interoperability.

[V2G20-2583] SECCID shall be created per C.3.

[V2G20-2584] SECCID shall be contained in the subject field of the SECC certificate as follows:

- the SECCID itself (see [V2G20-2583]) shall be the value of the Common Name (CN) of the Distinguished Name (DN);

- the name of the EVSE Operator shall be encoded in the field Organization (O) using a unique identifier chosen by the EVSE Operator, to identify this EVSE Operator;
- the X.500 distinguished name in the subject field shall not contain any further values.

B.5 Certificate provisioning service certificate profiles

B.5.1 Based on curves as defined by [V2G20-2674]

Table B.7 outlines the certificate installation (provisioning) service certificates based on curves as defined by [V2G20-2674].

Table B.7 — Certificate installation (provisioning) service certificates based on curves as defined by [V2G20-2674]

ISO 15118-20 certificate profiles		Certificate provisioning service		
		CPS sub-CA1 Sub	CPS sub-CA2 Sub	CPS leaf certificate Leaf
TbsCertificate	Version	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)
	SerialNumber	(Positive integer)	(Positive integer)	(Positive integer)
	Signature	AlgorithmIdentifier	x	x
		algorithm	x ecdsa-with-SHA512	x ecdsa-with-SHA512
		parameters	-	-
	Issuer	Country	(x)	(x)
		Organization	x	x
		Organization unit	(x)	(x)
		Common name	x	x
		Domain component	(x)	(x)
	Validity	Validity	x [up to CPS]	x [up to CPS]
		notBefore	x	x
	time		x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
		notAfter	x	x
	time		x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
	Subject	Country	(x)	x
		Organization	x	x
		Organization unit	(x)	(x)
		Common name	x	x
		Domain component	(x)	(x) & "CPS"
	SubjectPublicKeyInfo	AlgorithmIdentifier	x	x
		algorithm	x id-ecPublicKey	x id-ecPublicKey
		parameters	x ECParameters	x ECParameters
		namedCurve	x secp521r1	x secp521r1

ISO 15118-20 certificate profiles		Certificate provisioning service		
		CPS sub-CA1 Sub	CPS sub-CA2 Sub	CPS leaf certificate Leaf
	SubjectPublicKey	x (BIT STRING)	x (BIT STRING)	x (BIT STRING)
	IssuerUniqueID	-	-	-
	SubjectUniqueID	-	-	-
Extensions	AuthorityKeyIdentifier	x / c id-ce- authorityKeyIdentifier	x / c id-ce- authorityKeyIdentifier	x / c id-ce- authorityKeyIdentifier
	keyIdentifier	x (see [V2G20-1851])	x (see [V2G20-1851])	x (see [V2G20-1851])
	authorityCertIssuer	(x)	(x)	(x)
	authorityCertSerialNumber	(x) (Positive integer)	(x) (Positive integer)	(x) (Positive integer)
	SubjectKeyIdentifier	x / c id-ce-subjectKeyIdentifier	x / c id-ce-subjectKeyIdentifier	x / c id-ce-subjectKeyIdentifier
	keyIdentifier	x (see [V2G20-1852])	x (see [V2G20-1852])	x (see [V2G20-1852])
	KeyUsage	x / c id-ce-keyUsage	x / c id-ce-keyUsage	x / c id-ce-keyUsage
	digitalSignature	0/1	0/1	1
	nonRepudiation (contentCommitment)	0/1	0/1	0/1
	keyEncipherment	0/1	0/1	0/1
	dataEncipherment	0	0	0
	keyAgreement	0/1	0/1	0/1
	keyCertSign	1	1	0
	cRLSign	1	1	0
	encipherOnly	0	0	0
	decipherOnly	0	0	0
	ExtendedKeyUsage	-	-	-
	CertificatePolicies	(x) / nc	(x) / nc	(x) / nc
	BasicConstraints	x / c id-ce-basicConstraints	x / c id-ce-basicConstraints	x / c id-ce-basicConstraints
	CA	true	true	false
	PathLength	1	0	-
	CRLDistributionPoints	(x) / nc id-ce- cRLDistributionPoints	(x) / nc id-ce- cRLDistributionPoints	(x) / nc id-ce- cRLDistributionPoints
	distributionPoint	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)
	reasons	(x) (revocation reasons covered by the CRL)	(x) (revocation reasons covered by the CRL)	(x) (revocation reasons covered by the CRL)
	cRLIssuer	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)	(x) (authority that issues the CRL. If not present, defaults to Issuer name above)	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)
	AuthorityInfoAccess (OCSP)	(x) / nc id-pe-authorityInfoAccess	(x) / nc id-pe-authorityInfoAccess	(x) / nc id-pe-authorityInfoAccess
	accessMethod	x id-ad-ocsp	x id-ad-ocsp	x id-ad-ocsp
	accessLocation	x (location of the OCSP responder)	x (location of the OCSP responder)	x (location of the OCSP responder)
SignatureAlgorithm	AlgorithmIdentifier	x	x	x
	algorithm	x ecdsa-with-SHA512	x ecdsa-with-SHA512	x ecdsa-with-SHA512
	parameters	-	-	-
SignatureValue		(Octet-String)	(Octet-String)	(Octet-String)

NOTE 1 ECPParameters is simply a container that contains the parameter values for the selected algorithm.

NOTE 2 OID values for "ecdsa-with-SHA512", "id-ecPublicKey" and "secp521r1" can be found in B.1.

B.5.2 Based on curves as defined by [V2G20-2319]

Table B.8 outlines the certificate installation (provisioning) service certificates based on curves as defined by [V2G20-2319].

Table B.8 — Certificate installation (provisioning) service certificates based on curves as defined by [V2G20-2319]

ISO 15118-20 certificate profiles		Certificate provisioning service		
		CPS sub-CA1 Sub	CPS sub-CA2 Sub	CPS leaf certificate Leaf
TbsCertificate	Version	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)
	SerialNumber	(Positive integer)	(Positive integer)	(Positive integer)
	Signature	AlgorithmIdentifier	x	x
		algorithm	x id-Ed448	x id-Ed448
		parameters	-	-
	Issuer	Country	(x)	(x)
		Organization	x	x
		Organization unit	(x)	(x)
		Common name	x	x
	Validity	Domain component	(x)	(x)
		Validity	x [up to CPS]	x [up to CPS]
		notBefore	x	x
			x	x
		time	(GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	(GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
		notAfter	x	x
			x	x
		time	(GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	(GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
	Subject	Country	(x)	x
		Organization	x	x
		Organization unit	(x)	(x)
		Common name	x	x
	SubjectPublic KeyInfo	Domain component	(x)	(x) & "CPS"
		AlgorithmIdentifier	x	x
		algorithm	x id-Ed448	x id-Ed448
		parameters	-	-
	IssuerUniqueID	SubjectPublicKey	x (BIT STRING)	x (BIT STRING)
		IssuerUniqueID	-	-
	SubjectUniqueID	SubjectUniqueID	-	-
		AuthorityKeyIdentifier	x / c id-ce- authorityKeyIdentifier	x / c id-ce- authorityKeyIdentifier
		keyIdentifier	x (see [V2G20-1851])	x (see [V2G20-1851])
		authorityCertIssuer	(x)	(x)
		authorityCertSerialNumber	(x) (Positive integer)	(x) (Positive integer)
		SubjectKeyIdentifier	x / c id-ce-subjectKeyIdentifier	x / c id-ce-subjectKeyIdentifier
	Extensions	keyIdentifier	x (see [V2G20-1852])	x (see [V2G20-1852])

ISO 15118-20 certificate profiles		Certificate provisioning service		
		CPS sub-CA1 Sub	CPS sub-CA2 Sub	CPS leaf certificate Leaf
	KeyUsage	x / c id-ce-keyUsage	x / c id-ce-keyUsage	x / c id-ce-keyUsage
	digitalSignature	0/1	0/1	1
	nonRepudiation (contentCommitment)	0/1	0/1	0/1
	keyEncipherment	0/1	0/1	0/1
	dataEncipherment	0	0	0
	keyAgreement	0/1	0/1	0/1
	keyCertSign	1	1	0
	cRLSign	1	1	0
	encipherOnly	0	0	0
	decipherOnly	0	0	0
	ExtendedKeyUsage	-	-	-
	CertificatePolicies	(x) / nc	(x) / nc	(x) / nc
	BasicConstraints	x / c id-ce-basicConstraints	x / c id-ce-basicConstraints	x / c id-ce-basicConstraints
	CA	true	true	false
	PathLength	1	0	-
	CRLDistributionPoints	(x) / nc id-ce- cRLDistributionPoints	(x) / nc id-ce- cRLDistributionPoints	(x) / nc id-ce- cRLDistributionPoints
	distributionPoint	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)
	reasons	(x) (revocation reasons covered by the CRL)	(x) (revocation reasons covered by the CRL)	(x) (revocation reasons covered by the CRL)
	cRLIssuer	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)
	AuthorityInfoAccess (OCSP)	(x) / nc id-pe-authorityInfoAccess	(x) / nc id-pe-authorityInfoAccess	(x) / nc id-pe-authorityInfoAccess
	accessMethod	x id-ad-ocsp	x id-ad-ocsp	x id-ad-ocsp
	accessLocation	x (location of the OCSP responder)	x (location of the OCSP responder)	x (location of the OCSP responder)
SignatureAlgorithm	AlgorithmIdentifier	x	x	x
	algorithm	x id-Ed448	x id-Ed448	x id-Ed448
	parameters	-	-	-
SignatureValue		(Raw BIT STRING)	(Raw BIT STRING)	(Raw BIT STRING)

B.5.3 Common requirements for certificate provisioning service certificate profiles

As mentioned in Table B.7 and Table B.8, it is optional to include the "certificatePolicies" extension in any of the charge point operator certificates. If the "certificatePolicies" extension is included, it can be marked as non-critical so that the relying party can ignore this extension if it cannot process it. This is done to improve the interoperability.

[V2G20-2585] If the CPS sub-CA1 certificate or CPS sub-CA2 certificate or CPS leaf certificate contains both CRLDistributionPoints and AuthorityInfoAccess extensions, the relying party may process just one or both or none as per the local policy of the relying party.

NOTE Table B.7 and Table B.8 specify that both CRLDistributionPoints or AuthorityInfoAccess extensions are optional and non-critical to indicate that either of them can be included but that it is not mandatory for the receiver of these certificates to process these extensions. This requirement clarifies that if the issuer desires to include both of these, it is up to the relying party to decide which one to process, if any.

[V2G20-2586] The EVCC shall not process CRLDistributionPoints extension if it is present in CPS sub-CA1 certificate or CPS sub-CA2 certificate or CPS leaf certificate.

[V2G20-2587] The EVCC shall not process AuthorityInfoAccess extension if it is present in CPS sub-CA1 Certificate or CPS sub-CA2 certificate or CPS leaf certificate.

B.6 e-Mobility service provider certificate profiles

B.6.1 Based on curves as defined by [V2G20-2674]

Table B.9 outlines the e-Mobility service provider certificates based on curves as defined by [V2G20-2674].

Table B.9 — e-Mobility service provider certificates based on curves as defined by [V2G20-2674]

ISO 15118-20 certificate profiles		e-Mobility service provider			
		eMSP root CA Root	eMSP sub-CA1 Sub	eMSP sub-CA2 Sub/leaf	Contract certificate Leaf
TbsCertificate	Version	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)
	SerialNumber	(Positive integer)	(Positive integer)	(Positive integer)	(Positive integer)
	Signature	AlgorithmIdentifier	x	x	x
		algorithm	x ecdsa-with-SHA512	x ecdsa-with-SHA512	x ecdsa-with-SHA512
		parameters	-	-	-
	Issuer	Country	(x)	(x)	(x)
		Organization	x	x	x
		Organization unit	(x)	(x)	(x)
		Common name	x	x	x
		Domain component	(x)	(x)	(x)
	Validity	Validity	x [up to eMSP]	x [up to eMSP]	x [up to eMSP]
		notBefore	x	x	x
		time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
		notAfter	x	x	x
		time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
	Subject	Country	(x)	(x)	x
		Organization	x	x	x

ISO 15118-20 certificate profiles		e-Mobility service provider			
		eMSP root CA Root	eMSP sub-CA1 Sub	eMSP sub-CA2 Sub/leaf	Contract certificate Leaf
	Organization unit	(x)	(x)	(x)	(x)
	Common name	x	x	x	x EMAID (see [V2G20-2588])
	Domain component	(x)	(x)	(x)	(x)
	AlgorithmIdentifier	x	x	x	x
	algorithm	x id-ecPublicKey	x id-ecPublicKey	x id-ecPublicKey	x id-ecPublicKey
SubjectPublic KeyInfo	parameters	x ECParameters	x ECParameters	x ECParameters	x ECParameters
	namedCurve	x secp521r1	x secp521r1	x secp521r1	x secp521r1
	SubjectPublicKey	x (BIT STRING)	x (BIT STRING)	x (BIT STRING)	x (BIT STRING)
	IssuerUniqueID	-	-	-	-
	SubjectUniqueID	-	-	-	-
Extensions	AuthorityKeyIdentifier	-	x / c id-ce- authorityKeyIdentifier	x / c id-ce- authorityKeyIdentifier	x / c id-ce-authorityKeyIdentifier
	keyIdentifier	-	x (see [V2G20-1851])	x (see [V2G20-1851])	x (see [V2G20-1851])
	authorityCertIssuer	-	(x)	(x)	(x)
	authorityCertSerial Number	-	(x) (Positive integer)	(x) (Positive integer)	(x) (Positive integer)
	SubjectKeyIdentifier	x / c id-ce- subjectKeyIdentifier	x / c id-ce- subjectKeyIdentifier	x / c id-ce- subjectKeyIdentifier	x / c id-ce-subjectKeyIdentifier
	keyIdentifier	x (see [V2G20-1852])	x (see [V2G20-1852])	x (see [V2G20-1852])	x (see [V2G20-1852])
	KeyUsage	x / c id-ce- keyUsage	x / c id-ce- keyUsage	x / c id-ce- keyUsage	x / c id-ce-keyUsage
	digitalSignature	0/1	0/1	0/1	1
	nonRepudiation (contentCommitment)	0/1	0/1	0/1	0/1
	keyEncipherment	0/1	0/1	0/1	0/1
	dataEncipherment	0	0	0	0
	keyAgreement	0/1	0/1	0/1	0/1
	keyCertSign	1	1	1	0
	cRLSign	0/1	1	1	0
	encipherOnly	0	0	0	0
	decipherOnly	0	0	0	0
	ExtendedKeyUsage	-	-	-	-
	CertificatePolicies	-	(x) / nc	(x) / nc	(x) / nc
	BasicConstraints	x / c id-ce- basicConstraints	x / c id-ce- basicConstraints	x / c id-ce- basicConstraints	x / c id-ce-basicConstraints
	CA	true	true	true	false
	PathLength	-	1	0	-
	CRLDistributionPoints (CRL)	-	(x) / c id-ce- cRLDistributionPoints	(x) / c id-ce- cRLDistributionPoints	(x) / c id-ce-cRLDistributionPoints
	distributionPoint	-	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)

ISO 15118-20 certificate profiles		e-Mobility service provider			
		eMSP root CA Root	eMSP sub-CA1 Sub	eMSP sub-CA2 Sub/leaf	Contract certificate Leaf
	reasons	-	(x) (revocation reasons covered by the CRL)	(x) (revocation reasons covered by the CRL)	(x) (revocation reasons covered by the CRL)
	cRLIssuer	-	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)
	AuthorityInfoAccess (OCSP)	-	(x) / c id-pe-authorityInfoAccess	(x) / c id-pe-authorityInfoAccess	(x) / c id-pe-authorityInfoAccess
	accessMethod	-	x id-ad-ocsp	x id-ad-ocsp	x id-ad-ocsp
	accessLocation	-	x (location of the OCSP responder)	x (location of the OCSP responder)	x (location of the OCSP responder)
	SubjectInfoAccess (ContractMobilityOperatorName)	-	-	-	(x) / nc id-pe-subjectInfoAccess
	accessMethod	-	-	-	x id-contractOperatorName (see [V2G20-1226])
	accessLocation	-	-	-	x (A user-friendly name of the eMSP for direct display in the vehicle that issued the contract. Syntax: An alphanumeric string with a maximum length of 255 characters)
	SubjectInfoAccess (ContractTariffName)	-	-	-	(x) / nc id-pe-subjectInfoAccess
	accessMethod	-	-	-	x id-contractTariffName (see [V2G20-1226])
	accessLocation	-	-	-	x (A user-friendly name of the tariff for direct display in the vehicle. Syntax: An alphanumeric string with a maximum length of 255 characters)
	SubjectInfoAccess (ContractDynamicInformationUrl)	-	-	-	(x) / nc id-pe-subjectInfoAccess
	accessMethod	-	-	-	x id-contractDynamicInformationUrl (see [V2G20-1226])
	accessLocation	-	-	-	x (A URL to a distribution point of current (daily) additional information to the contract Syntax: An alphanumeric string with a maximum length of 255 characters)
SignatureAlgorithm	AlgorithmIdentifier	x	x	x	x
	algorithm	x ecdsa-with-SHA512	x ecdsa-with-SHA512	x ecdsa-with-SHA512	x ecdsa-with-SHA512

ISO 15118-20 certificate profiles		e-Mobility service provider			
		eMSP root CA Root	eMSP sub-CA1 Sub	eMSP sub-CA2 Sub/leaf	Contract certificate Leaf
	parameters	-	-	-	-
SignatureValue		(Octet-String)	(Octet-String)	(Octet-String)	(Octet-String)

NOTE 1 ECParameters is simply a container that contains the parameter values for the selected algorithm.

NOTE 2 OID values for "ecdsa-with-SHA512", "id-ecPublicKey" and "secp521r1" can be found in B.1.

NOTE 3 IETF RFC 5280 has been updated by IETF RFC 6818, IETF RFC 8398 and IETF RFC 8399. These updates are considered to be included in this document.

B.6.2 Based on curves as defined by [V2G20-2319]

Table B.10 outlines the e-Mobility service provider certificates based on curves as defined by [V2G20-2319].

Table B.10 — e-Mobility service provider certificates based on curves as defined by [V2G20-2319]

ISO 15118-20 certificate profiles			e-Mobility service provider			
			eMSP root CA Root	eMSP sub-CA1 Sub	eMSP sub-CA2 Sub/leaf	Contract certificate Leaf
TbsCertificate	Version		2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)
	SerialNumber		(Positive integer)	(Positive integer)	(Positive integer)	(Positive integer)
	Signature	AlgorithmIdentifier	x	x	x	x
		algorithm	id-Ed448	id-Ed448	id-Ed448	id-Ed448
		parameters	-	-	-	-
	Issuer	Country	(x)	(x)	(x)	(x)
		Organization	x	x	x	x
		Organization unit	(x)	(x)	(x)	(x)
		Common name	x	x	x	x
		Domain component	(x)	(x)	(x)	(x)
	Validity	Validity	x [up to eMSP]	x [up to eMSP]	x [up to eMSP]	x [up to eMSP]
		notBefore	x	x	x	x
		time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
		notAfter	x	x	x	x

ISO 15118-20 certificate profiles		e-Mobility service provider			
		eMSP root CA Root	eMSP sub-CA1 Sub	eMSP sub-CA2 Sub/leaf	Contract certificate Leaf
	time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
	Subject	Country	(x)	(x)	(x)
		Organization	x	x	x
		Organization unit	(x)	(x)	(x)
		Common name	x	x	x EMAID (see [V2G20-2588])
		Domain component	(x)	(x)	(x)
	SubjectPublic KeyInfo	AlgorithmIdentifier	x	x	x
		algorithm	x id-Ed448	x id-Ed448	x id-Ed448
		parameters-	-	-	-
		SubjectPublicKey	x (BIT STRING)	x (BIT STRING)	x (BIT STRING)
	IssuerUniqueID		-	-	-
	SubjectUniqueID		-	-	-
	Extensions	AuthorityKeyIdentifier	-	x / c id-ce- authorityKeyIdentifier	x / c id-ce-authorityKeyIdentifier
		keyIdentifier	-	x (see [V2G20-1851])	x (see [V2G20-1851])
		authorityCertIssuer	-	(x)	(x)
		authorityCertSerial Number	-	(x) (Positive integer)	(x) (Positive integer)
		SubjectKeyIdentifier	x / c id-ce- subjectKeyIdentifier	x / c id-ce- subjectKeyIdentifier	x / c id-ce-subjectKeyIdentifier
		keyIdentifier	x (see [V2G20-1852])	x (see [V2G20-1852])	x (see [V2G20-1852])
		KeyUsage	x / c id-ce- keyUsage	x / c id-ce- keyUsage	x / c id-ce-keyUsage
		digitalSignature	0/1	0/1	1
		nonRepudiation (contentCommitment)	0/1	0/1	0/1
		keyEncipherment	0/1	0/1	0/1
		dataEncipherment	0	0	0
		keyAgreement	0/1	0/1	0/1
		keyCertSign	1	1	0
		cRLSign	0/1	1	0
		encipherOnly	0	0	0
		decipherOnly	0	0	0
		ExtendedKeyUsage	-	-	-
		CertificatePolicies	-	(x) / nc	(x) / nc
		BasicConstraints	x / c id-ce- basicConstraints	x / c id-ce- basicConstraints	x / c id-ce-basicConstraints
		CA	true	true	false

ISO 15118-20 certificate profiles		e-Mobility service provider			
		eMSP root CA Root	eMSP sub-CA1 Sub	eMSP sub-CA2 Sub/leaf	Contract certificate Leaf
	PathLength	-	1	0	-
	CRLDistributionPoints (CRL)	-	(x) / c id-ce- cRLDistributionPoints	(x) / c id-ce- cRLDistributionPoints	(x) / c id-ce-cRLDistributionPoints
	distributionPoint	-	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)
	reasons	-	(x) (revocation reasons covered by the CRL)	(x) (revocation reasons covered by the CRL)	(x) (revocation reasons covered by the CRL)
	cRLIssuer	-	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)
	AuthorityInfoAccess (OCSP)	-	(x) / c id-pe- authorityInfoAccess	(x) / c id-pe- authorityInfoAccess	(x) / c id-pe-authorityInfoAccess
	accessMethod	-	x id-ad-ocsp	x id-ad-ocsp	x id-ad-ocsp
	accessLocation	-	x (location of the OCSP responder)	x (location of the OCSP responder)	x (location of the OCSP responder)
	SubjectInfoAccess (ContractMobilityOperatorName)	-	-	-	(x) / nc id-pe-subjectInfoAccess
	accessMethod	-	-	-	x id-contractOperatorName (see [V2G20-1226])
	accessLocation	-	-	-	x (A user-friendly name of the eMSP for direct display in the vehicle that issued the contract. Syntax: An alphanumeric string with a maximum length of 255 characters)
	SubjectInfoAccess (ContractTariffName)	-	-	-	(x) / nc id-pe-subjectInfoAccess
	accessMethod	-	-	-	x id-contractTariffName (see [V2G20-1226])
	accessLocation	-	-	-	(x) (A user-friendly name of the tariff for direct display in the vehicle. Syntax: An alphanumeric string with a maximum length of 255 characters)
	SubjectInfoAccess (ContractDynamicInformationUrl)	-	-	-	(x) / nc id-pe-subjectInfoAccess
	accessMethod	-	-	-	x id-contractDynamicInformationUrl (see [V2G20-1226])

ISO 15118-20 certificate profiles			e-Mobility service provider			
			eMSP root CA Root	eMSP sub-CA1 Sub	eMSP sub-CA2 Sub/leaf	Contract certificate Leaf
		accessLocation	-	-	-	x (A URL to a distribution point of current (daily) additional information to the contract. Syntax: An alphanumeric string with a maximum length of 255 characters)
SignatureAlgorithm	AlgorithmIdentifier		x	x	x	x
	algorithm		x id-Ed448	x id-Ed448	x id-Ed448	x id-Ed448
	parameters		-	-	-	-
SignatureValue			(Raw BIT STRING)	(Raw BIT STRING)	(Raw BIT STRING)	(Raw BIT STRING)

NOTE IETF RFC 5280 has been updated by IETF RFC 6818, IETF RFC 8398 and IETF RFC 8399. These updates are considered to be included in this document.

B.6.3 Common requirements for e-mobility service provider certificate profiles

eMSP sub-CA2 has two purposes:

- it certifies contract certificates;

AND

- it signs tariff information destined for the owners of those contract certificates.

The ID of the provisioning certificate (PCID) is used by the eMSP to identify the contract that belongs to this EV. This is possible because the customer has given the PCID (CN and O, see L.4 and its subclauses for details) to the eMSP when creating the contract. For this purpose, PCID is supposed to be a short string that is unique to the OEM that created the provisioning certificate and to the vehicle which is requesting the contract certificate(s). Furthermore, it is contained in the provisioning certificate.

As mentioned in Table B.9 and Table B.10, it is optional to include the "certificatePolicies" extension in any of the e-Mobility service provider certificates. If the "certificatePolicies" extension is included, it can be marked as non-critical so that the relying party can ignore this extension if it cannot process it. This is done to improve the interoperability.

It should be noted that the EVCC does not validate the contract certificate or any other certificates included in that chain. Usually, the SECC also does not validate the contract certificate or any other certificates included in that chain. Usually, the EVCC provides the contract certificate chain to the eMSP during AuthorizationReq. customarily, it is up to the eMSP to validate the contract certificate chain and ensure that it meets the requirements. In rare instances, the SECC itself may validate the contract certificate chain for authorization purposes.

[V2G20-2588] EMAID shall be created per C.1.

[V2G20-2589] EMAID shall be contained in the subject field of the contract certificate as follows:

- the EMAID itself (see [V2G20-2588]) shall be the value of the Common Name (CN) of the Distinguished Name (DN);
- the name of the eMSP shall be encoded in the field Organization (O) using a unique identifier chosen by the eMSP, to identify this eMSP;
- the X.500 distinguished name in the subject field shall not contain any further values.

[V2G20-2590] eMSP sub-CA1 certificate, eMSP sub-CA2 certificate and contract certificate shall contain at least one or both of CRLDistributionPoints or/and AuthorityInfoAccess extensions.

NOTE 1 Table B.9 and Table B.10 specify that both CRLDistributionPoints and AuthorityInfoAccess extensions are optional but critical to indicate that either one of them are included and processed. [V2G20-2590] clarifies that including at least one of them is mandatory while including both of them is not mandatory.

NOTE 2 [V2G20-2590] does not specify which one of these extensions can be included. It is left up to the eMSP to decide which certificate revocation method works best for it. In some cases CRLs will work better as they allow the relying party to download the entire CRL once and have it available for local checking of revocation status of the next contract certificate without a need for a live connection to the CRL server while in other cases OCSP will provide better results as it provides current revocation status and requires less storage and searching on part of the relying party.

[V2G20-2591] If eMSP sub-CA1 certificate or eMSP sub-CA2 certificate or contract certificate contains both CRLDistributionPoints and AuthorityInfoAccess extensions, the relying party may process just one or both as per the local policy of the relying party.

NOTE 3 Table B.9 and Table B.10 specify that both CRLDistributionPoints and AuthorityInfoAccess extensions are optional but critical to indicate that at least one of them are included and that whichever one is included is processed. [V2G20-2591] clarifies that if the issuer desires to include both of these, it is up to the relying party to decide which one to process. However, the relying party processes at least one.

NOTE 4 Per ITU-T X.509 and IETF RFC 5280, if any of these extensions are present and critical, they are processed during certificate validation.

NOTE 5 IETF RFC 5280 has been updated by IETF RFC 6818, IETF RFC 8398 and IETF RFC 8399. These updates are considered to be included in this document.

[V2G20-1225] The contract certificate extensions in Table B.9 and Table B.10 shall use the definition of "SubjectInfoAccess" as defined IETF RFC 5280:2008, 4.2.2.2, so that the issuer of a contract certificate can provide the applicant with additional information and services. As mentioned in IETF RFC 5280, the field "accessMethod" indicates the type² of information and the field "accessLocation" indicates either the information itself or a storage location for the information.

NOTE 6 IETF RFC 5280 has been updated by IETF RFC 6818, IETF RFC 8398 and IETF RFC 8399. These updates are considered to be included in this document.

[V2G20-1226] The issuer of the certificate shall describe the information with three additional access descriptions for "AccessDescription". Here are the definitions in ASN.1 notation:

² OIDs see <https://www.oid-info.com>

ISO 15118-20:2022(E)

```
ISO15118-Extensions { iso(1) standard(0) 15118 part20(20) extensions(0) }
DEFINITIONS :=
BEGIN

id-15118-20-ad                                OBJECT IDENTIFIER := { iso(1) standard(0) 15118 part20(20)
                                                extensions(0) }

-- a user-friendly name of the subject's eMobility Service Provider
id-contractOperatorName                      OBJECT IDENTIFIER := { id-15118-20-ad
                                                contractOperatorName(1) }

-- a user-friendly name of the subject's tariff
id-contractTariffName                        OBJECT IDENTIFIER := { id-15118-20-ad
                                                contractTariffName(2) }

-- the URL to a distribution point of (daily) current additional information of the subject's contract
id-contractDynamicInformationUrl             OBJECT IDENTIFIER := { id-15118-20-ad
                                                contractDynamicInformationUrl(3) }

END
```

NOTE 7 Refer to ITU-T X.680 for details of ASN.1 notation.

ContractDynamicInformationUrl provides a URL to a distribution point of current (daily) additional information to the contract for the customer/EV. This contract information may be downloaded and processed by the EV (or any other IT system managing the contract for the customer/EV). EV may use any methods available to it to download this data. This includes, the internet services (VAS) provided by the EVSE, or any other method available to the EV to reach the provided URL. The usage of the data located at the URL is up to the OEM's implementation.

B.7 OEM provisioning certificate profiles

B.7.1 Based on curves as defined by [V2G20-2674]

Table B.11 outlines OEM provisioning certificates based on curves as defined by [V2G20-2674].

Table B.11 — OEM provisioning certificates based on curves as defined by [V2G20-2674]

ISO 15118-2 certificate profiles		EV manufacturer: OEM provisioning			
		OEM root CA Root	OEM sub-CA1 Sub	OEM sub-CA2 Sub	OEM provisioning certificate Leafs
TbsCertificate	Version	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)
	SerialNumber	(Positive integer)	(Positive integer)	(Positive integer)	(Positive integer)
	Signature	AlgorithmIdentifier	x	x	x
		algorithm	x ecdsa-with-SHA512	x ecdsa-with-SHA512	x ecdsa-with-SHA512
		parameters	-	-	-
	Issuer	Country	(x)	(x)	(x)
		Organization	x	x	x
		Organization unit	(x)	(x)	(x)
		Common name	x	x	x
		Domain component	(x)	(x)	(x)
	Validity	Validity	x [up to OEM]	x [up to OEM]	x [up to OEM]
		notBefore	x	x	x

ISO 15118-2 certificate profiles		EV manufacturer: OEM provisioning			
		OEM root CA Root	OEM sub-CA1 Sub	OEM sub-CA2 Sub	OEM provisioning certificate Leafs
	time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
	notAfter	x	x	x	x
	time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
Subject	Country	(x)	(x)	(x)	(x)
	Organization	x	x	x	x
	Organization unit	(x)	(x)	(x)	(x)
	Common name	x	x	x	x PCID (see [V2G20-1780])
	Domain component	(x)	(x)	(x)	(x) (shall not end with "EV")
SubjectPublic KeyInfo	AlgorithmIdentifier	x	x	x	x
	algorithm	x id-ecPublicKey	x id-ecPublicKey	x id-ecPublicKey	x id-ecPublicKey
	parameters	x ECParameters	x ECParameters	x ECParameters	x ECParameters
	namedCurve	x secp521r1	x secp521r1	x secp521r1	x secp521r1
	SubjectPublicKey	x (BIT STRING)	x (BIT STRING)	x (BIT STRING)	x (BIT STRING)
IssuerUniqueID		-	-	-	-
SubjectUniqueID		-	-	-	-
Extensions	AuthorityKeyIdentifier	-	x / c id-ce- authorityKeyId entifier	x / c id-ce- authorityKeyId entifier	x / c id-ce-authorityKeyIdentifier
	keyIdentifier	-	x (see [V2G20-1851])	x (see [V2G20-1851])	x (see [V2G20-1851])
	authorityCertIssuer	-	(x)	(x)	(x)
	authorityCertSerial Number	-	(x) (Positive integer)	(x) (Positive integer)	(x) (Positive integer)
	SubjectKeyIdentifier	x / c id-ce- subjectKeyId entifier	x / c id-ce- subjectKeyId entifier	x / c id-ce- subjectKeyId entifier	x / c id-ce-subjectKeyIdentifier
	keyIdentifier	x (see [V2G20-1852])	x (see [V2G20-1852])	x (see [V2G20-1852])	x (see [V2G20-1852])
	KeyUsage	x / c id-ce-keyUsage	x / c id-ce-keyUsage	x / c id-ce-keyUsage	x / c id-ce-keyUsage
	digitalSignature	0/1	0/1	0/1	1

ISO 15118-2 certificate profiles		EV manufacturer: OEM provisioning			
		OEM root CA Root	OEM sub-CA1 Sub	OEM sub-CA2 Sub	OEM provisioning certificate Leafs
	nonRepudiation (contentCommitment)	0/1	0/1	0/1	0/1
	keyEncipherment	0/1	0/1	0/1	0/1
	dataEncipherment	0	0	0	0
	keyAgreement	0/1	0/1	0/1	1
	keyCertSign	1	1	1	0
	cRLSign	0/1	1	1	0
	encipherOnly	0	0	0	0
	decipherOnly	0	0	0	0
	ExtendedKeyUsage	-	-	-	-
	CertificatePolicies	-	(x) / nc	(x) / nc	(x) / nc
	BasicConstraints	x / c id-ce-basicConstraints	x / c id-ce-basicConstraints	x / c id-ce-basicConstraints	x / c id-ce-basicConstraints
	CA	true	true	true	false
	PathLength	-	1	0	-
	CRLDistributionPoints (CRL)	-	(x) / c id-ce-cRLDistributionPoints	(x) / c id-ce-cRLDistributionPoints	(x) / c id-ce-cRLDistributionPoints
	distributionPoint	-	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)
	reasons	-	(x) (revocation reasons covered by the CRL)	(x) (revocation reasons covered by the CRL)	(x) (revocation reasons covered by the CRL)
	cRLIssuer	-	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)
	AuthorityInfoAccess (OCSP)	-	(x) / c id-pe-authorityInfoAccess	(x) / c id-pe-authorityInfoAccess	(x) / c id-pe-authorityInfoAccess
	accessMethod	-	x id-ad-ocsp	x id-ad-ocsp	x id-ad-ocsp
	accessLocation	-	x (location of the OCSP responder)	x (location of the OCSP responder)	x (location of the OCSP responder)
	SubjectInfoAccess (TpmStorageKey)	-	-	-	(x) / nc id-pe-subjectInfoAccess
	accessMethod	-	-	-	x id-tpmStorageKey (see [V2G20-2596])
	accessLocation	-	-	-	x (EVCC's TPM public storage key that can be used by the issuing SA to encrypt the newly generated private key so that this information can be stored directly in the EVCC's TPM. Syntax: 132 byte public key in base64 encoding)
	SubjectInfoAccess (TpmPolicyDigest)	-	-	-	(x) / nc id-pe-subjectInfoAccess

ISO 15118-2 certificate profiles		EV manufacturer: OEM provisioning			
		OEM root CA Root	OEM sub-CA1 Sub	OEM sub-CA2 Sub	OEM provisioning certificate Leafs
	accessMethod	-	-	-	x id-tpmPolicyDigest (see [V2G20-2596])
	accessLocation	-	-	-	x (EVCC's TPM policy digest that shall be added to the information generated by the issuing SA before encryption. Syntax: 64 byte hash value in base64 encoding)
SignatureAlgorithm	AlgorithmIdentifier	x	x	x	x
	algorithm	x ecdsa-with-SHA512	x ecdsa-with-SHA512	x ecdsa-with-SHA512	x ecdsa-with-SHA512
	parameters	-	-	-	-
SignatureValue		(Octet-String)	(Octet-String)	(Octet-String)	(Octet-String)

NOTE 1 ECPParameters is simply a container that contains the parameter values for the selected algorithm.

NOTE 2 OID values for "ecdsa-with-SHA512", "id-ecPublicKey" and "secp521r1" can be found in B.1.

B.7.2 Based on curves as defined by [V2G20-2319]

Table B.12 outlines OEM provisioning certificates based on curves as defined by [V2G20-2319].

Table B.12 — OEM provisioning certificates based on curves as defined by [V2G20-2319]

ISO 15118-2 certificate profiles		EV manufacturer: OEM provisioning			
		OEM root CA Root	OEM sub-CA1 Sub	OEM sub-CA2 Sub	OEM provisioning certificate Leafs
TbsCertificate	Version	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)
	SerialNumber	(Positive integer)	(Positive integer)	(Positive integer)	(Positive integer)
	Signature	AlgorithmIdentifier	x	x	x
		algorithm	x id-Ed448	x id-Ed448	x id-Ed448
		parameters	-	-	-
	Issuer	Country	(x)	(x)	(x)
		Organization	x	x	x
		Organization unit	(x)	(x)	(x)
		Common name	x	x	x
		Domain component	(x)	(x)	(x)
	Validity	Validity	x [up to OEM]	x [up to OEM]	x [up to OEM]
		notBefore	x	x	x
		time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
		notAfter	x	x	x

ISO 15118-2 certificate profiles		EV manufacturer: OEM provisioning			
		OEM root CA Root	OEM sub-CA1 Sub	OEM sub-CA2 Sub	OEM provisioning certificate Leafs
	time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
Subject	Country	(x)	(x)	(x)	(x)
	Organization	x	x	x	x
	Organization unit	(x)	(x)	(x)	(x)
	Common name	x	x	x	x PCID (see [V2G20-1780])
	Domain component	(x)	(x)	(x)	(x) (shall not end with "EV")
SubjectPublic KeyInfo	AlgorithmIdentifier	x	x	x	x
	algorithm	x id-Ed448	x id-Ed448	x id-Ed448	x id-Ed448
	parameters	-	-	-	-
	SubjectPublicKey	x / c (BIT STRING)	x / c (BIT STRING)	x / c (BIT STRING)	x / c (BIT STRING)
IssuerUniqueID		-	-	-	-
SubjectUniqueID		-	-	-	-
Extensions	AuthorityKeyIdentifier	-	x / c id-ce- authorityKeyId entifier	x / c id-ce- authorityKeyId entifier	x / c id-ce-authorityKeyIdentifier
	keyIdentifier	-	x (see [V2G20-1851])	x (see [V2G20-1851])	x (see [V2G20-1851])
	authorityCertIssuer	-	(x)	(x)	(x)
	authorityCertSerial Number	-	(x) (Positive integer)	(x) (Positive integer)	(x) (Positive integer)
	SubjectKeyIdentifier	x / c id-ce- subjectKeyId entifier	x / c id-ce- subjectKeyId entifier	x / c id-ce- subjectKeyId entifier	x / c id-ce-subjectKeyIdentifier
	keyIdentifier	x (see [V2G20-1852])	x (see [V2G20-1852])	x (see [V2G20-1852])	x (see [V2G20-1852])
	KeyUsage	x / c id-ce-keyUsage	x / c id-ce-keyUsage	x / c id-ce-keyUsage	x / c id-ce-keyUsage
	digitalSignature	0/1	0/1	0/1	1
	nonRepudiation (contentCommitment)	0/1	0/1	0/1	0/1
	keyEncipherment	0/1	0/1	0/1	0/1
	dataEncipherment	0	0	0	0
	keyAgreement	0/1	0/1	0/1	1
	keyCertSign	1	1	1	0
	cRLSign	0/1	1	1	0
	encipherOnly	0	0	0	0
	decipherOnly	0	0	0	0
	ExtendedKeyUsage	-	-	-	-
	CertificatePolicies	-	(x) / nc	(x) / nc	(x) / nc
	BasicConstraints	x / c id-ce- basicConstraints	x / c id-ce- basicConstraints	x / c id-ce- basicConstraints	x / c id-ce-basicConstraints
	CA	true	true	true	false
	PathLength	-	1	0	-

ISO 15118-2 certificate profiles		EV manufacturer: OEM provisioning			
		OEM root CA Root	OEM sub-CA1 Sub	OEM sub-CA2 Sub	OEM provisioning certificate Leafs
	CRLDistributionPoints (CRL)	-	(x) / c id-ce- cRLDistributionPoints	(x) / c id-ce- cRLDistributionPoints	(x) / c id-ce-cRLDistributionPoints
	distributionPoint	-	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)
	reasons	-	(x) (revocation reasons covered by the CRL)	(x) (revocation reasons covered by the CRL)	(x) (revocation reasons covered by the CRL)
	cRLIssuer	-	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)
	AuthorityInfoAccess (OCSP)	-	(x) / c id-pe- authorityInfoAccess	(x) / c id-pe- authorityInfoAccess	(x) / c id-pe-authorityInfoAccess
	accessMethod	-	x id-ad-ocsp	x id-ad-ocsp	x id-ad-ocsp
	accessLocation	-	x (location of the OCSP responder)	x (location of the OCSP responder)	x (location of the OCSP responder)
	SubjectInfoAccess (TpmStorageKey)	-	-	-	(x) / nc id-pe-subjectInfoAccess
	accessMethod	-	-	-	x id-tpmStorageKey (see [V2G20-2596])
	accessLocation	-	-	-	x (EVCC's TPM public storage key that can be used by the issuing SA to encrypt the newly generated private key so that this information can be stored directly in the EVCC's TPM. Syntax: 132 byte public key in base64 encoding)
	SubjectInfoAccess (TpmPolicyDigest)	-	-	-	(x) / nc id-pe-subjectInfoAccess
	accessMethod	-	-	-	x id-tpmPolicyDigest (see [V2G20-2596])
	accessLocation	-	-	-	x (EVCC's TPM policy digest that shall be added to the information generated by the issuing SA before encryption. Syntax: 64 byte hash value in base64 encoding)
	SubjectAltPublicKeyInfo	-	-	-	x / c id-ce-subjectAltPublicKeyInfo
	algorithmIdentifier	-	-	-	x
	algorithm	-	-	-	x id-X448
	parameters	-	-	-	-

ISO 15118-2 certificate profiles			EV manufacturer: OEM provisioning			
			OEM root CA Root	OEM sub-CA1 Sub	OEM sub-CA2 Sub	OEM provisioning certificate Leafs
		subjectAltPublicKey	-	-	-	x (BIT STRING)
SignatureAlgorithm	AlgorithmIdentifier		x	x	x	x
	algorithm		x id-Ed448	x id-Ed448	x id-Ed448	x id-Ed448
	parameters		-	-	-	-
SignatureValue			(Raw BIT STRING)	(Raw BIT STRING)	(Raw BIT STRING)	(Raw BIT STRING)

B.7.3 Common requirements for OEM provisioning certificate profiles

As mentioned in Table B.11 and Table B.12 it is optional to include the "certificatePolicies" extension in any of the OEM provisioning certificates. If the "certificatePolicies" extension is included, it can be marked as non-critical so that the relying party can ignore this extension if it cannot process it. This is done to improve the interoperability.

It should be noted that neither EVCC nor SECC validate the OEM provisioning certificate or any other certificates included in that chain. The EVCC provides this certificate to the eMSP during CertificateInstallationReq. It is up to the eMSP to validate the certificate and ensure it meets the requirements.

When the configurable mechanism as defined by [V2G20-2320] indicates that the curve as defined by [V2G20-2319] needs to be used, the EVCC will send its OEM provisioning certificate conforming to Table B.11 to the eMSP via the SECC. When the configurable mechanism as defined by [V2G20-2320] indicates that the curve as defined by [V2G20-2674] needs to be used, the EVCC will send its OEM provisioning certificate conforming to Table B.12 to the eMSP via the SECC. When the eMSP receives an OEM provisioning certificate conforming to Table B.12, it will use SubjectPublicKey defined in Table B.11 for ECDHE per 7.9.2.5.4. When the eMSP receives an OEM provisioning certificate conforming to Table B.12, it will use subjectAltPublicKey defined in Table B.12 for ECDHE per 7.9.2.5.4.

[V2G20-1780] PCID shall be created per C.2.

[V2G20-1781] PCID shall be contained in the subject field of the provisioning certificate as follows:

- the PCID itself (see [V2G20-1780]) shall be the value of the common name (CN) of the distinguished name (DN);
- the name of the OEM shall be encoded in the field organization (O) using a unique identifier chosen by the OEM, to identify this OEM;
- the X.500 distinguished name in the subject field shall not contain any further values.

[V2G20-2592] OEM sub-CA1 certificate, OEM sub-CA2 certificate and OEM provisioning certificate shall contain at least one or both of CRLDistributionPoints or/and AuthorityInfoAccess extensions.

NOTE 1 Table B.11 and Table B.12 specify that both CRLDistributionPoints and AuthorityInfoAccess extensions are optional but critical to indicate that at least one is included and that it is processed. [V2G20-2592] clarifies that including at least one of them is mandatory while including both of them is not mandatory.

NOTE 2 [V2G20-2592] does not specify which one of these extensions can be included. It is left up to the OEM to decide which certificate revocation method works best for it. In some cases CRLs will work better as they allow the relying party to download the entire CRL once and have it available for local checking of revocation status of the next contract certificate without a need for a live connection to the CRL server while in other cases OCSP will provide better results as it provides current revocation status and requires less storage and searching on part of the relying party.

[V2G20-2593] If OEM sub-CA1 certificate or OEM sub-CA2 certificate or OEM provisioning certificate contains both CRLDistributionPoints and AuthorityInfoAccess extensions, the relying party may process just one or both as per the local policy of the relying party. The relying party, though, shall process at least one of these extensions.

NOTE 3 Table B.11 and Table B.12 specify that both CRLDistributionPoints and AuthorityInfoAccess extensions are optional but critical to indicate that at least one of them is included and that whichever one is included is processed. [V2G20-2593] clarifies that if the issuer desires to include both of these, it's up to the relying party to decide which one to process. However, the relying party processes at least one.

NOTE 4 Per IETF RFC 5280, if any of these extensions are present and critical, they are processed during certificate validation.

NOTE 5 IETF RFC 5280 has been updated by IETF RFC 6818, IETF RFC 8398 and IETF RFC 8399. These updates are considered to be included in this document.

[V2G20-2594] The OEM provisioning certificate extensions in Table B.11 and Table B.12 shall use the definition of "SubjectInfoAccess" as defined in IETF 5280, 4.2.2.2, so that the issuer of an OEM provisioning certificate can indicate to the eMSP that the EVCC contains a TPM and provide the necessary information to the eMSP to process the contract certificates for TPM. As mentioned in IETF RFC 5280, the field "accessMethod" indicates the type³ of information and the field "accessLocation" indicates either the information itself or a storage location for the information.

NOTE 6 IETF RFC 5280 has been updated by IETF RFC 6818, IETF RFC 8398 and IETF RFC 8399. These updates are considered to be included in this document.

[V2G20-2595] In case the OEM provisioning certificate provided by/for the EVCC contains the non-critical extension "TPMStorageKey" and/or "TpmPolicyDigest", the SA/eMSP shall use the method described in 7.9.2.5.3 to encrypt the private contract certificate key for this EVCC.

[V2G20-2596] The issuer of the certificate shall describe the information with three additional access descriptions for "AccessDescription". Here are the definitions in ASN.1 notation:

```
ISO15118-Extensions { iso(1) standard(0) 15118 part20(20) extensions(0) }
DEFINITIONS :=
BEGIN
id-15118-20-ad                                OBJECT IDENTIFIER ::= { iso(1) standard(0) 15118 part20(20)
                                                extensions(0) }

-- Public Storage Key of EVCC's TPM
id-tpmStorageKey                               OBJECT IDENTIFIER ::= { id-15118-20-ad tpmStorageKey(4) }

-- Policy Digest for EVCC's TPM Contract Keys
id-tpmPolicyDigest                             OBJECT IDENTIFIER ::= { id-15118-20-ad tpmPolicyDigest (5) }
END
```

NOTE 7 Refer to ITU-T X.680 for details of ASN.1 notation.

B.8 Vehicle certificate profiles

³ OIDs see <https://www.oid-info.com>

B.8.1 Based on curves as defined by [V2G20-2674]

Table B.13 outlines the vehicle certificates based on curves as defined by [V2G20-2674].

Table B.13 — Vehicle certificates based on curves as defined by [V2G20-2674]

ISO 15118-2 certificate profiles		EV manufacturer: vehicle certificates		
		Vehicle sub-CA1 Sub	Vehicle sub-CA2 Sub	Vehicle certificate Leaf
TbsCertificate	Version	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)
	SerialNumber	(Positive integer)	(Positive integer)	(Positive integer)
	Signature	AlgorithmIdentifier	x	x
		algorithm	x ecdsa-with-SHA512	x ecdsa-with-SHA512
		parameters	-	-
	Issuer	Country	(x)	(x)
		Organization	x	x
		Organization unit	(x)	(x)
		Common name	x	x
		Domain component	(x)	(x)
	Validity	Validity	x [up to OEM]	x [up to OEM]
		notBefore	x	x
		time	x	x
			(GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ)	(GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ)
			[Actual time is CA discretionary]	[Actual time is CA discretionary]
		notAfter	x	x
		time	x	x
			(GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ)	(GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ)
			[Actual time is CA discretionary]	[Actual time is CA discretionary]
	Subject	Country	(x)	(x)
		Organization	x	x
		Organization unit	(x)	(x)
		Common name	x	x EVCCID (see [V2G20-2597])
		Domain component	(x)	(x) & "EV"
	SubjectPublicKeyInfo	AlgorithmIdentifier	x	x
		algorithm	x id-ecPublicKey	x id-ecPublicKey
		parameters	x ECParameters	x ECParameters
		namedCurve	x secp521r1	x secp521r1
		SubjectPublicKey	x	x
			(BIT STRING)	(BIT STRING)
	IssuerUniqueID	-	-	-
	SubjectUniqueID	-	-	-
	Extensions	AuthorityKeyIdentifier	x / c id-ce- authorityKeyIdentifier	x / c id-ce- authorityKeyIdentifier
		keyIdentifier	x (see [V2G20-1851])	x (see [V2G20-1851])
		authorityCertIssuer	(x)	(x)
		authorityCertSerial Number	(x) (Positive integer)	(x) (Positive integer)

ISO 15118-2 certificate profiles		EV manufacturer: vehicle certificates		
		Vehicle sub-CA1 Sub	Vehicle sub-CA2 Sub	Vehicle certificate Leaf
		x / c id-ce-subjectKeyIdentifier	x / c id-ce-subjectKeyIdentifier	x / c id-ce-subjectKeyIdentifier
	SubjectKeyIdentifier	x / c id-ce-subjectKeyIdentifier	x / c id-ce-subjectKeyIdentifier	x / c id-ce-subjectKeyIdentifier
	keyIdentifier	x (see [V2G20-1852])	x (see [V2G20-1852])	x (see [V2G20-1852])
	KeyUsage	x / c id-ce-keyUsage	x / c id-ce-keyUsage	x / c id-ce-keyUsage
	digitalSignature	0/1	0/1	1
	nonRepudiation (contentCommitment)	0/1	0/1	0/1
	keyEncipherment	0/1	0/1	0/1
	dataEncipherment	0	0	0
	keyAgreement	0/1	0/1	0/1
	keyCertSign	1	1	0
	cRLSign	1	1	0
	encipherOnly	0	0	0
	decipherOnly	0	0	0
	ExtendedKeyUsage	-	-	(x) / c id-ce-extKeyUsage
	keyPurposeId	-	-	x id-kp-clientAuth
	keyPurposeId	-	-	(x) id-kp-serverAuth
	CertificatePolicies	(x) / nc	(x) / nc	(x) / nc
	BasicConstraints	x / c id-ce-basicConstraints	x / c id-ce-basicConstraints	x / c id-ce-basicConstraints
	CA	true	true	false
	PathLength	1	0	-
	CRLDistributionPoints (CRL)	(x) / c id-ce- cRLDistributionPoints	(x) / c id-ce-cRLDistributionPoints	(x) / c id-ce- cRLDistributionPoints
	distributionPoint	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)
	reasons	(x) (revocation reasons covered by the CRL)	(x) (revocation reasons covered by the CRL)	(x) (revocation reasons covered by the CRL)
	cRLIssuer	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)
	AuthorityInfoAccess (OCSP)	(x) / c id-pe-authorityInfoAccess	(x) / c id-pe-authorityInfoAccess	(x) / c id-pe-authorityInfoAccess
	accessMethod	x id-ad-ocsp	x id-ad-ocsp	x id-ad-ocsp
	accessLocation	x (location of the OCSP responder)	x (location of the OCSP responder)	x (location of the OCSP responder)
SignatureAlgorithm	AlgorithmIdentifier	x	x	x
	algorithm	x ecdsa-with-SHA512	x ecdsa-with-SHA512	x ecdsa-with-SHA512
	parameters	-	-	-
SignatureValue		(Octet-String)	(Octet-String)	(Octet-String)

NOTE 1 ECPParameters is simply a container that contains the parameter values for the selected algorithm.

NOTE 2 OID values for "ecdsa-with-SHA512", "id-ecPublicKey" and "secp521r1" can be found in B.1.

Although including "keyPurposeId" (as part of "ExtendedKeyUsage") is mentioned as optional, it is highly recommended to include this parameter in the vehicle certificate. This makes the certificate more secure and prevents its misuse.

B.8.2 Based on curves as defined by [V2G20-2319]

Table B.14 outlines the vehicle certificates based on curves as defined by [V2G20-2319].

Table B.14 — Vehicle certificates based on curves as defined by [V2G20-2319]

ISO 15118-2 certificate profiles		EV manufacturer: vehicle certificates		
		Vehicle sub-CA1 Sub	Vehicle sub-CA2 Sub	Vehicle certificate Leaf
TbsCertificate	Version	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)
	SerialNumber	(Positive integer)	(Positive integer)	(Positive integer)
	Signature	AlgorithmIdentifier	x	x
		algorithm	x id-Ed448	x id-Ed448
		parameters	-	-
	Issuer	Country	(x)	(x)
		Organization	x	x
		Organization unit	(x)	(x)
		Common name	x	x
	Validity	Domain component	(x)	(x)
		Validity	x [up to OEM]	x [up to OEM]
		notBefore	x	x
		time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
		notAfter	x	x
		time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
	Subject	Country	(x)	(x)
		Organization	x	x
		Organization unit	(x)	(x)
		Common name	x	x EVCCID (see [V2G20-2597])
		Domain component	(x)	(x) & "EV"
	SubjectPublicKeyInfo	AlgorithmIdentifier	x	x
		algorithm	x id-Ed448	x id-Ed448
		SubjectPublicKey	x (BIT STRING)	x (BIT STRING)
	IssuerUniqueID		-	-
	SubjectUniqueID		-	-
	Extensions	AuthorityKeyIdentifier	x / c id-ce- authorityKeyIdentifier	x / c id-ce- authorityKeyIdentifier
		keyIdentifier	x (see [V2G20-1851])	x (see [V2G20-1851])
		authorityCertIssuer	(x)	(x)
		authorityCertSerial Number	(x) (Positive integer)	(x) (Positive integer)
		SubjectKeyIdentifier	x / c id-ce-subjectKeyIdentifier	x / c id-ce-subjectKeyIdentifier
		keyIdentifier	x (see [V2G20-1852])	x (see [V2G20-1852])
		KeyUsage	x / c id-ce-keyUsage	x / c id-ce-keyUsage
		digitalSignature	0/1	1

ISO 15118-2 certificate profiles		EV manufacturer: vehicle certificates		
		Vehicle sub-CA1 Sub	Vehicle sub-CA2 Sub	Vehicle certificate Leaf
	nonRepudiation (contentCommitment)	0/1	0/1	0/1
	keyEncipherment	0/1	0/1	0/1
	dataEncipherment	0	0	0
	keyAgreement	0/1	0/1	0/1
	keyCertSign	1	1	0
	cRLSign	1	1	0
	encipherOnly	0	0	0
	decipherOnly	0	0	0
	ExtendedKeyUsage	-	-	(x) / c id-ce-extKeyUsage
	keyPurposeId	-	-	x id-kp-clientAuth
	keyPurposeId	-	-	(x) id-kp-serverAuth
	CertificatePolicies	(x) / nc	(x) / nc	(x) / nc
	BasicConstraints	x / c id-ce-basicConstraints	x / c id-ce-basicConstraints	x / c id-ce-basicConstraints
	CA	true	true	false
	PathLength	1	0	-
	CRLDistributionPoints (CRL)	(x) / c id-ce- cRLDistributionPoints	(x) / c id-ce-cRLDistributionPoints	(x) / c id-ce- cRLDistributionPoints
	distributionPoint	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)
	reasons	(x) (revocation reasons covered by the CRL)	(x) (revocation reasons covered by the CRL)	(x) (revocation reasons covered by the CRL)
	cRLIssuer	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)
	AuthorityInfoAccess (OCSP)	(x) / c id-pe-authorityInfoAccess	(x) / c id-pe-authorityInfoAccess	(x) / c id-pe-authorityInfoAccess
	accessMethod	x id-ad-ocsp	x id-ad-ocsp	x id-ad-ocsp
	accessLocation	x (location of the OCSP responder)	x (location of the OCSP responder)	x (location of the OCSP responder)
SignatureAlgorithm	AlgorithmIdentifier	x	x	x
	algorithm	x id-Ed448	x id-Ed448	x id-Ed448
	parameters	-	-	-
SignatureValue		(Raw BIT STRING)	(Raw BIT STRING)	(Raw BIT STRING)

Although including "keyPurposeId" (as part of "ExtendedKeyUsage") is mentioned as optional, it is highly recommended to include this parameter in the vehicle certificate. This makes the certificate more secure and prevents its misuse.

B.8.3 Common requirements for vehicle certificate profiles

As mentioned in Table B.13 and Table B.14, it is optional to include the "certificatePolicies" extension in any of the vehicle certificates. If the "certificatePolicies" extension is included, it can be marked as non-critical so that the relying party can ignore this extension if it cannot process it. This is done to improve the interoperability.

[V2G20-2597] EVCCID shall be created per C.5.

[V2G20-2598] EVCCID shall be contained in the subject field of the vehicle certificate as follows:

- the EVCCID itself (see [V2G20-2597]) shall be the value of the Common Name (CN) of the Distinguished Name (DN);
- the name of the OEM shall be encoded in the field Organization (O) using a unique identifier chosen by the OEM, to identify this OEM;
- the X.500 distinguished name in the subject field shall not contain any further values.

[V2G20-2599] Vehicle sub-CA1 certificate, vehicle sub-CA2 certificate and vehicle certificate shall contain at least one or both of CRLDistributionPoints or/and AuthorityInfoAccess extensions.

NOTE 1 Table B.13 and Table B.14 specify that both CRLDistributionPoints and AuthorityInfoAccess extensions are optional but critical to indicate that one of them is included and processed. [V2G20-2599] clarifies that including at least one of them is mandatory while including both of them is not mandatory.

NOTE 2 [V2G20-2599] does not specify which one of these extensions can be included. It is left up to the OEM to decide which certificate revocation method works best for it. In some cases CRLs will work better as they allow the relying party to download the entire CRL once and have it available for local checking of revocation status of the next contract certificate without a need for a live connection to the CRL server while in other cases OCSP will provide better results as it provides current revocation status and requires less storage and searching on part of the relying party.

[V2G20-2600] If vehicle sub-CA1 certificate, vehicle sub-CA2 certificate and vehicle certificate contains both CRLDistributionPoints and AuthorityInfoAccess extensions, the relying party may process just one or both as per the local policy of the relying party. The relying party, though, shall process at least one of these extensions.

NOTE 3 Table B.13 and Table B.14 specify that both CRLDistributionPoints and AuthorityInfoAccess extensions are optional but critical to indicate that at least one of them is included and that whichever one is included is processed. [V2G20-2600] clarifies that if the issuer desires to include both of these, it is up to the relying party to decide which one to process. However, the relying party processes at least one.

NOTE 4 Per ITU-T X.509 and IETF RFC 5289, if any of these extensions are present and critical, they are processed during certificate validation.

NOTE 5 IETF RFC 5280 has been updated by IETF RFC 6818, IETF RFC 8398 and IETF RFC 8399. These updates are considered to be included in this document.

There may be use cases allowing usage of vehicle certificate for both client and server authentication. As such, the vehicle certificate profile provides for the possibility of allowing the vehicle certificate to be used for both. Since these use cases are not fully defined yet, it is left up to the OEM's discretion whether to generate vehicle certificates that can be used for both client and server authentication or those that can only be used for client authentication.

To allow widest acceptance of the vehicle certificate, vehicle certificates should be derived from a chain originating from a V2G root CA certificate. This allows the SECC to validate the vehicle certificate using a V2G root that it may possess. If the vehicle certificates cannot be derived from a chain originating from a V2G root CA certificate, the vehicle certificate chain may be cross signed by a V2G root CA certificate. Refer to H.1.5 and its subclauses for further details on cross-signing.

B.9 Cross certificate & OCSP signer certificate profiles

B.9.1 Based on curves as defined by [V2G20-2674]

Table B.15 outlines the cross certificate and OCSP signer certificates based on curves as defined by [V2G20-2674].

Table B.15 — Cross certificate and OSCP signer certificate based on curves as defined by [V2G20-2674]

ISO 15118-20 certificate profiles			Cross cross certificate	OCSP OCSP signer certificate
TbsCertificate	Version		2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)
	SerialNumber		(Positive integer)	(Positive integer)
	Signature	AlgorithmIdentifier	x	x
		algorithm	x ecdsa-with-SHA512	x ecdsa-with-SHA512
		parameters	-	-
	Issuer	Country	(x)	(x)
		Organization	x	x
		Organization unit	(x)	(x)
		Common name	x	x
	Validity	Domain component	(x)	(x)
		Validity	x [up to issuer]	x [up to 1 year]
		notBefore	x	x
		time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
		notAfter	x	x
		time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
	Subject	Country	(x)	-
		Organization	x	x
		Organization unit	(x)	(x)
		Common name	x	x
		Domain component	(x)	(x)
	SubjectPublicKey Info	AlgorithmIdentifier	x	x
		algorithm	x id-ecPublicKey	x id-ecPublicKey
		parameters	x ECParameters	x ECParameters
		namedCurve	x secp521r1	x secp521r1
		SubjectPublicKey	x (BIT STRING)	x (BIT STRING)
	IssuerUniqueID		-	-
	SubjectUniqueID		-	-
	Extensions	AuthorityKeyIdentifier	x / nc id-ce-authorityKeyIdentifier	x / nc id-ce-authorityKeyIdentifier
		keyIdentifier	x (see [V2G20-1851])	x (see [V2G20-1851])
		authorityCertIssuer	(x)	(x)
		authorityCertSerialNumber	(x) (Positive integer)	(x) (Positive integer)
		SubjectKeyIdentifier	x / nc id-ce-subjectKeyIdentifier	x / nc id-ce-subjectKeyIdentifier
		keyIdentifier	x (see [V2G20-1852])	x (see [V2G20-1852])
		KeyUsage	x / c id-ce-keyUsage	x / c id-ce-keyUsage
		digitalSignature	0/1	0/1
		nonRepudiation (contentCommitment)	0/1	0/1
		keyEncipherment	0/1	0/1
		dataEncipherment	0	0

ISO 15118-20 certificate profiles			Cross cross certificate	OCSP OCSP signer certificate
		keyAgreement	0/1	0/1
		keyCertSign	1	0
		cRLSign	0/1	0
		encipherOnly	0	0
		decipherOnly	0	0
		ExtendedKeyUsage	-	x / c id-ce-extKeyUsage
		keyPurposeId	-	x id-kp-OCSPSigning
		CertificatePolicies	(x) / nc	-
		BasicConstraints	x / c id-ce-basicConstraints	x / c id-ce-basicConstraints
		CA	true	false
		PathLength	-	-
		CRLDistributionPoints (CRL)	(x) / c id-ce-cRLDistributionPoints	-
		distributionPoint	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)	-
		reasons	(x) (revocation reasons covered by the CRL)	-
		cRLIssuer	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)	-
		AuthorityInfoAccess (OCSP)	(x) / c id-pe-authorityInfoAccess	-
		accessMethod	x id-ad-ocsp	-
		accessLocation	x (location of the OCSP responder)	-
		SubjectInfoAccess (CrossCertIndication)	x / nc id-pe-subjectInfoAccess	-
		accessMethod	x id-crossCertIndication (see [V2G20-2604])	-
		accessLocation	x "CROSS" (Syntax: An alphanumeric string with a maximum length of 5 characters as described here)	-
SignatureAlgorit hm	AlgorithmIdentifier		x	x
	algorithm		x ecdsa-with-SHA512	x ecdsa-with-SHA512
	parameters		-	-
SignatureValue			(Octet-String)	(Octet-String)
			(This cross certificate profile shall be applied for all cross certificates regardless of the certificate structure.)	(This OCSP signer certificate profile shall be applied for all sub- CAs and leafs that use OCSP regardless of the certificate structure.)

NOTE 1 ECPParameters is simply a container that contains the parameter values for the selected algorithm.

NOTE 2 OID values for "ecdsa-with-SHA512", "id-ecPublicKey" and "secp521r1" can be found in B.1.

B.9.2 Based on curves as defined by [V2G20-2319]

Table B.16 outlines the cross certificate and OCSP signer certificates based on curves as defined by [V2G20-2319].

Table B.16 — Cross certificate and OCSP signer certificate based on curves as defined by [V2G20-2319]

ISO 15118-20 certificate profiles			Cross cross certificate	OCSP OCSP signer certificate
TbsCertificate	Version		2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)
	SerialNumber		(Positive integer)	(Positive integer)
	Signature	AlgorithmIdentifier	x	x
		algorithm	x id-Ed448	x id-Ed448
	Issuer	Country	(x)	(x)
		Organization	x	x
		Organization Unit	(x)	(x)
		Common Name	x	x
		Domain Component	(x)	(x)
	Validity	Validity	x [up to issuer]	x [up to 1 year]
		notBefore	x	x
		time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
		notAfter	x	x
		time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
	Subject	Country	(x)	-
		Organization	x	x
		Organization unit	(x)	(x)
		Common name	x	x
		Domain component	(x)	(x)
	SubjectPublicKey Info	AlgorithmIdentifier	x	x
		algorithm	x id-Ed448	x id-Ed448
		SubjectPublicKey	x (BIT STRING)	x (BIT STRING)
	IssuerUniqueID		-	-
	SubjectUniqueID		-	-
	Extensions	AuthorityKeyIdentifier	x / c id-ce-authorityKeyIdentifier	x / c id-ce-authorityKeyIdentifier
		keyIdentifier	x (see [V2G20-1851])	x (see [V2G20-1851])
		authorityCertIssuer	(x)	(x)
		authorityCertSerialNum ber	(x) (Positive integer)	(x) (Positive integer)
		SubjectKeyIdentifier	x / c id-ce-subjectKeyIdentifier	x / c id-ce-subjectKeyIdentifier
		keyIdentifier	x (see [V2G20-1852])	x (see [V2G20-1852])
		KeyUsage	x / c	x / c
		digitalSignature	0/1	0/1
		nonRepudiation (contentCommitment)	0/1	0/1
		keyEncipherment	0/1	0/1
		dataEncipherment	0	0
		keyAgreement	0/1	0/1
		keyCertSign	1	0
		cRLSign	0/1	0
		encipherOnly	0	0
		decipherOnly	0	0
		ExtendedKeyUsage	-	x / c id-ce-extKeyUsage
		keyPurposeId	-	x id-kp-OCSPSigning

ISO 15118-20 certificate profiles			Cross cross certificate	OCSP OCSP signer certificate
		CertificatePolicies	(x) / nc	-
		BasicConstraints	x / c id-ce-basicConstraints	x / c id-ce-basicConstraints
		CA	true	false
		PathLength	-	-
		CRLDistributionPoints (CRL)	x / c id-ce-cRLDistributionPoints	-
		distributionPoint	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)	-
		reasons	(x) (revocation reasons covered by the CRL)	-
		cRLIssuer	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)	-
		AuthorityInfoAccess (OCSP)	x / c id-pe-authorityInfoAccess	-
		accessMethod	x id-ad-ocsp	-
		accessLocation	x (location of the OCSP responder)	-
		SubjectInfoAccess (CrossCertIndication)	x / nc id-pe-subjectInfoAccess	-
		accessMethod	x id-crossCertIndication (see [V2G20-2604])	-
		accessLocation	x "CROSS" (Syntax: An alphanumeric string with a maximum length of 5 characters as described here)	-
SignatureAlgorit hm	AlgorithmIdentifier		x	x
	algorithm		x id-Ed448	x id-Ed448
	parameters		-	-
SignatureValue			(Raw BIT STRING)	(Raw BIT STRING)
			(This cross certificate profile shall be applied for all cross certificates regardless of the certificate structure.)	(This OCSP signer certificate profile shall be applied for all sub- CAs and leafs that use OCSP regardless of the certificate structure.)

B.9.3 Common requirements for cross certificate & OCSF signer certificate profiles

As mentioned in Table B.15 and Table B.16, it is optional to include the "certificatePolicies" extension in any of the cross certificates. If the "certificatePolicies" extension is included, it can be marked as non-critical so that the relying party can ignore this extension if it cannot process it. This is done to improve the interoperability.

[V2G20-2601] Cross certificate shall contain at least one or both of CRLDistributionPoints or/and AuthorityInfoAccess extensions.

NOTE 1 Table B.15 and Table B.16 specify that both CRLDistributionPoints and AuthorityInfoAccess extensions are optional but critical to indicate that one of them is included and processed. [V2G20-2601] clarifies that including at least one of them is mandatory while including both of them is not mandatory.

NOTE 2 [V2G20-2601] does not specify which one of these extensions can be included. It is left up to the CA to decide which certificate revocation method works best for it. In some cases CRLs will work better as they allow the relying party to download the entire CRL once and have it available for local checking of revocation status of the next contract certificate without a need for a live connection to the CRL server while in other cases OCSF will provide better results as it provides current revocation status and requires less storage and searching on part of the relying party.

[V2G20-2602] If cross certificate contains both CRLDistributionPoints and AuthorityInfoAccess extensions, the relying party may process just one or both as per the local policy of the relying party. The relying party, though, shall process at least one of these extensions.

NOTE 3 Table B.15 and Table B.16 specify that both CRLDistributionPoints and AuthorityInfoAccess extensions are required and critical to indicate that at least one of them is included and that whichever one is included is processed. [V2G20-2602] clarifies that if the issuer desires to include both of these, it is up to the relying party to decide which one to process. However, the relying party processes at least one.

NOTE 4 Per ITU-T X.509 and IETF RFC 5280, if any of these extensions are present and critical, they are processed during certificate validation.

NOTE 5 IETF 5280 has been updated by IETF RFC 6818, IETF RFC 8398 and IETF RFC 8399. These updates are considered to be included in this document.

[V2G20-2603] The cross certificate extensions in Table B.15 and Table B.16 shall use the definition of "SubjectInfoAccess" as defined in IETF 5280, 4.2.2.2, to indicate that the particular certificate is cross certified. As mentioned in IETF RFC 5280, the field "accessMethod" indicates the type⁴ of information and the field "accessLocation" indicates either the information itself or a storage location for the information.

NOTE 6 IETF RFC 5280 has been updated by IETF RFC 6818, IETF RFC 8398 and IETF RFC 8399. These updates are considered to be included in this document.

The relying party is not required to take any action based on the cross certificate extension "SubjectInfoAccess". It is simply provided for informational purposes.

[V2G20-2604] The issuer of the certificate shall describe the information with three additional access descriptions for "AccessDescription". Here are the definitions in ASN.1 notation:

```
ISO15118-Extensions { iso(1) standard(0) 15118 part20(20) extensions(0) }
DEFINITIONS :=
BEGIN

id-15118-20-ad                                OBJECT IDENTIFIER ::= { iso(1) standard(0) 15118 part20(20)
                                                extensions(0) }

-- Indication that this certificate is a cross-certificate
id-crossCertIndication                        OBJECT IDENTIFIER ::= { id-15118-20-ad
                                                crossCertIndication (6) }

END
```

NOTE 7 Refer to ITU-T X.680 for details of ASN.1 notation.

Refer to H.1.5 and its subclauses for further details on cross-signing.

B.10 Private environment certificate profiles

B.10.1 Based on curves as defined by [V2G20-2674]

Table B.17 outlines certificates for a private environment based on curves as defined by [V2G20-2674].

Table B.17 — Certificates for a private environment based on curves as defined by [V2G20-2674]

⁴ OIDs see <https://www.oid-info.com>

ISO 15118-2 certificate profiles		Private SECC/private environment			
		PE private root CA Root	PE sub-CA1 Sub (optional)	PE sub-CA2 Sub (optional)	PE certification Leaf
TbsCertificate	Version	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)
	SerialNumber	(Positive integer)	(Positive integer)	(Positive integer)	(Positive integer)
	Signature	AlgorithmIdentifier	x	x	x
		algorithm	x ecdsa-with-SHA512	x ecdsa-with-SHA512	x ecdsa-with-SHA512
		parameters	-	-	-
	Issuer	Country	(x)	(x)	(x)
		Organization	x	x	x
		Organization unit	(x)	(x)	(x)
		Common name	x	x	x
	Validity	Domain component	(x)	(x)	(x)
		Validity	x [40 years]	x [up to PE]	x [up to PE]
		notBefore	x	x	x
		time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
		notAfter	x	x	x
		time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
		Subject	(x)	(x)	(x)
	SubjectPublic KeyInfo	Country	(x)	(x)	x
		Organization	x	x	x
		Organization unit	(x)	(x)	(x)
		Common name	x	x	x PEID (see [V2G20-2697])
		Domain component	(x)	(x)	(x)
	SubjectPublic KeyInfo	AlgorithmIdentifier	x	x	x
		algorithm	x id-ecPublicKey	x id-ecPublicKey	x id-ecPublicKey
		parameters	x ECParameters	x ECParameters	x ECParameters
		namedCurve	x secp521r1	x secp521r1	x secp521r1
		SubjectPublicKey	x (BIT STRING)	x (BIT STRING)	x (BIT STRING)
	IssuerUniqueID		-	-	-
	SubjectUniqueID		-	-	-
	Extensions	AuthorityKeyIdentifier	-	x / c id-ce- authorityKeyIdentifier	x / c id-ce- authorityKeyIdentifier
		keyIdentifier	-	x (see [V2G20-1851])	x (see [V2G20-1851])

ISO 15118-2 certificate profiles		Private SECC/private environment			
		PE private root CA Root	PE sub-CA1 Sub (optional)	PE sub-CA2 Sub (optional)	PE certification Leaf
	authorityCertIssuer	-	(x)	(x)	(x)
	authorityCertSerial Number	-	(x) (Positive integer)	(x) (Positive integer)	(x) (Positive integer)
	SubjectKeyIdentifier	x / c id-ce- subjectKeyIdentifier	x / c id-ce- subjectKeyIdentifier	x / c id-ce- subjectKeyIdentifier	x / c id-ce- subjectKeyIdentifier
	keyIdentifier	x (see [V2G20-1852])	x (see [V2G20-1852])	x (see [V2G20-1852])	x (see [V2G20-1852])
	KeyUsage	x / c id-ce-keyUsage	x / c id-ce-keyUsage	x / c id-ce-keyUsage	x / c id-ce-keyUsage
	digitalSignature	0/1	0/1	0/1	1
	nonRepudiation (contentCommitment)	0/1	0/1	0/1	0/1
	keyEncipherment	0/1	0/1	0/1	0/1
	dataEncipherment	0	0	0	0
	keyAgreement	0/1	0/1	0/1	0/1
	keyCertSign	1	1	1	0
	cRLSign	0/1	1	1	0
	encipherOnly	0	0	0	0
	decipherOnly	0	0	0	0
	ExtendedKeyUsage	-	-	-	(x) / c id-ce-extKeyUsage
	keyPurposeId	-	-	-	x id-kp-serverAuth
	CertificatePolicies	-	(x) / nc	(x) / nc	(x) / nc
	BasicConstraints	x / c id-ce- basicConstraints	x / c id-ce- basicConstraints	x / c id-ce- basicConstraints	x / c id-ce- basicConstraints
	CA	true	true	true	false
	PathLength	-	1	0	-
	CRLDistributionPoints (CRL)	-	(x) / c id-ce- cRLDistributionPoints	(x) / c id-ce- cRLDistributionPoints	(x) / c id-ce- cRLDistributionPoints
	distributionPoint	-	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)	(x) (location of the CRL distributor, if not present, defaults to cRLIssuer)
	reasons	-	(x) (revocation reasons covered by the CRL)	(x) (revocation reasons covered by the CRL)	(x) (revocation reasons covered by the CRL)
	cRLIssuer	-	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)	(x) (authority that issues the CRL, if not present, defaults to Issuer name above)
	AuthorityInfoAccess (OCSP)	-	(x) / c id-pe- authorityInfoAccesses	(x) / c id-pe- authorityInfoAccesses	(x) / c id-pe- authorityInfoAccesses
	accessMethod	-	x id-ad-ocsp	x id-ad-ocsp	x id-ad-ocsp
	accessLocation	-	x (location of the OCSP responder)	x (location of the OCSP responder)	x (location of the OCSP responder)
SignatureAlgorithm	AlgorithmIdentifier	x	x	x	x
	algorithm	x ecdsa-with-SHA512	x ecdsa-with-SHA512	x ecdsa-with-SHA512	x ecdsa-with-SHA512
	parameters	-	-	-	-
SignatureValue		(Octet-String)	(Octet-String)	(Octet-String)	(Octet-String)

NOTE 1 ECPParameters is simply a container that contains the parameter values for the selected algorithm.

NOTE 2 OID values for "ecdsa-with-SHA512", "id-ecPublicKey" and "secp521r1" can be found in B.1.

B.10.2 Based on curves as defined by [V2G20-2319]

Table B.18 outlines certificates for a private environment based on curves as defined by [V2G20-2319].

Table B.18 — Certificates for a private environment based on curves as defined by [V2G20-2319]

ISO 15118-2 certificate profiles		Private SECC/private environment			
		PE private root CA Root	PE sub-CA1 Sub (optional)	PE sub-CA2 Sub (optional)	PE certification Leaf
TbsCertificate	Version	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)	2 ("2" indicates X.509v3)
	SerialNumber	(Positive integer)	(Positive integer)	(Positive integer)	(Positive integer)
	Signature	AlgorithmIdentifier	x	x	x
		algorithm	x id-Ed448	x id-Ed448	x id-Ed448
		parameters	-	-	-
	Issuer	Country	(x)	(x)	(x)
		Organization	x	x	x
		Organization unit	(x)	(x)	(x)
		Common name	x	x	x
	Validity	Domain component	(x)	(x)	(x)
		Validity	x [40 years]	x [up to PE]	x [up to PE]
		notBefore	x	x	x
		time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
		notAfter	x	x	x
		time	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]	x (GeneralizedTime expressed in Greenwich Mean Time (Zulu) with format YYYYMMDDHHMMSSZ) [Actual time is CA discretionary]
	Subject	Country	(x)	(x)	x
		Organization	x	x	x
		Organization unit	(x)	(x)	(x)
		Common name	x	x	x PEID (see [V2G20-2697])
		Domain component	(x)	(x)	(x)
	SubjectPublicKeyInfo	AlgorithmIdentifier	x	x	x
		algorithm	x id-Ed448	x id-Ed448	x id-Ed448
		SubjectPublicKey	x (BIT STRING)	x (BIT STRING)	x (BIT STRING)
	IssuerUniqueID		-	-	-
	SubjectUniqueID		-	-	-